DESIGN AND DEVELOPMENT OF IOT-BASED SMART MALDIVE FISH PROCESSING SYSTEM

INDIVIDUAL THESIS

BANDARA C.S

IT22340214

Individual component: Intelligent IoT-Based Multi-Parameter Environmental Monitoring for Fish Processing Optimization

Group ID: R26-IT-028

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISO6R CERTIFICATION

Candidate declaration. I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. I have disclosed all use of generative AI and other automated tools in accordance with the module policy. I remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this thesis.

Student name: Bandara C.S

Student ID: IT22340214

Signature: _____ Date: _____

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Mr. Supunya Swarnakantha		
Co-supervisor: Ms. Dulani Jayasinghe		

ABSTRACT

Traditional Maldive fish drying processes are highly influenced by environmental conditions, particularly temperature, relative humidity, airflow, and the presence of gases associated with spoilage. In many processing environments, these conditions are observed manually, making it difficult to continuously detect environmental changes that may negatively affect drying efficiency and product quality. To address this limitation, this research developed an intelligent IoI-based multi-parameter environmental monitoring system for fish processing optimization as part of a wider smart Maldive fish processing system. The individual component was designed to continuously monitor the drying environment using an ESP32-based IoT node integrated with temperature and humidity sensing, airflow measurement, and mq-135 gas sensing. Sensor readings were collected at regular intervals, filtered to reduce short-term fluctuations, and transmitted through MQTT for real-time monitoring. A threshold-based decision mechanism was also implemented to identify unsafe environmental conditions and generate timely alerts when undesirable conditions persisted. During a continuous seven-day dehydration cycle, the system recorded 10,080 validated one-minute telemetry packets and achieved a 99.6% MQTT packet delivery rate. During a rainfall-related environmental disturbance, the system detected the unsafe drying condition and generated an alert within 2.1 seconds. The monitored drying process achieved a stabilized dehydration duration of 4.2 ± 0.3 days, compared with 6.5 ± 1.4 days under unmonitored ambient conditions. These results indicate that the developed system can provide reliable real-time environmental monitoring and support timely corrective actions during fish drying. The individual contribution of this research is the design, implementation, and validation of a multi-parameter IoT-based environmental monitoring and alerting subsystem specifically developed for Maldive fish processing.

Keywords: internet of things, environmental monitoring, fish drying, ESP32, real-time monitoring

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Mr. Supunya Swarnakantha, for his continuous guidance, valuable feedback, and support throughout the research and development of the Intelligent IoT-based multi-parameter environmental monitoring system for fish processing optimization. His academic advice and technical direction were important in shaping the research methodology, system design, implementation, and evaluation of this individual component.

I am also deeply grateful to my co-supervisor, Ms. Dulani Jayasinghe, for her constructive suggestions, encouragement, and guidance throughout the project. Her feedback helped improve both the technical quality of the developed system and the academic presentation of this research.

I would also like to thank the academic staff of the faculty of computing at the Sri Lanka institute of information technology for providing the knowledge, resources, and learning environment required to complete this research successfully. I am grateful to my research group members for their cooperation and contribution to the overall IoT-based smart Maldive fish processing system, particularly during component integration and system-level development.

Finally, i would like to express my heartfelt appreciation to my family and friends for their patience, encouragement, and continuous support throughout my academic journey. Their motivation and understanding helped me remain focused and committed during the research, development, testing, and documentation stages of this thesis.

TABLE OF CONTENTS

1 INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 2

1.3 Problem statement 3

1.4 Research gap 4

1.5 Aim and objectives 5

1.6 Research questions or hypotheses 6

1.7 Scope limitations and assumptions 7

1.8 Individual contribution 8

1.9 Thesis structure 9

2 LITERATURE REVIEW AND RELATED WORK 11

2.1 Review method and scope 11

2.2 Domain and theoretical foundations 12

2.3 Related research 13

2.4 Existing systems products and techniques 14

2.5 Gap synthesis and conceptual framework 16

3 RESEARCH METHODOLOGY 18

3.1 Research design 18

3.2 Data participants or study materials 19

3.3 Experimental or study procedure 20

3.4 Measures and evaluation metrics 21

3.5 Data analysis 22

3.6 Ethics privacy security and responsible AI 23
3.7 Reproducibility and research artefacts 24
4 PRODUCT AND SOLUTION DESIGN 25
4.1 Requirements 25
4.2 Component architecture within the complete solution 26
4.3 Design alternatives and decisions 27
4.4 Detailed design 28
4.5 Standards and professional practice 29
5 IMPLEMENTATION INTEGRATION AND TESTING 31
5.1 Development environment and technology stack 31
5.2 Component implementation 32
5.3 Integration with the other components 33
5.4 Verification and software testing 34
5.5 Deployment and operation 35
6 RESULTS AND VALIDATION 37
6.1 Research results 37
6.2 Product and technical results 37
6.3 User or stakeholder validation 39
6.4 Integration results 39
6.5 Summary of findings 40
7 DISCUSSION 41
7.1 Interpretation of findings 41
7.2 Comparison with literature baselines and existing solutions 42
7.3 Research contribution 44
7.4 Product contribution 45
7.5 Limitations and threats to validity 46

7.6 Implications 47
8 CONCLUSION AND FUTURE DIRECTIONS 49
8.1 Conclusion 49
8.2 Achievement of Objectives 50
8.3 Individual Research and Product Contributions 51
8.4 Future Directions 52
REFERENCES 55
GLOSSARY 57
APPENDICES 59
Appendix A Research Artefacts and Extended Technical Evidence 59
Appendix B Academic Integrity and AI Use Disclosure 60

LIST OF FIGURES

Figure 1:Conceptual Framework of the Intelligent IoT-Based Multi-Parameter Environmental Monitoring System 17
Figure 2:Research Methodology Process of the IoT-Based Environmental Monitoring Study 21
Figure 3:Detailed Architecture of the IoT-Based Multi-Parameter Environmental Monitoring Component 29
Figure 4:Implementation, Integration and Testing Architecture of the IoT-Based Environmental Monitoring Component 36
Figure 5:Rainfall event humidity + airflow graph 38

LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
DHT22	Digital Humidity and Temperature Sensor
DSR	Design Science Research
ESP32	Espressif 32-bit Microcontroller
IoT	Internet of Things
MCU	Microcontroller Unit
MQTT	Message Queuing Telemetry Transport
MOS	Metal-Oxide Semiconductor
NH_3	Ammonia
RH	Relative Humidity
TMA	Trimethylamine
TVB-N	Total Volatile Basic Nitrogen
VOC	Volatile Organic Compound
Wi-Fi	Wireless Fidelity

1 INTRODUCTION

1.1 Overall project context

Maldive fish, locally known as *Umbalakada*, is a traditional dried fish product that forms an important part of Sri Lankan food culture and small-scale coastal food processing. The production process involves several interconnected stages, including raw fish selection, cleaning, boiling, salting, drying, and final storage. The quality achieved at one stage can directly influence the effectiveness of the next stage, making consistent process control important throughout the complete production cycle. In conventional processing environments, many of these activities depend heavily on manual observation and the practical experience of operators. This can lead to variation in processing conditions, especially when environmental factors change unexpectedly.

To address these limitations, the overall research project, Design and Development of an IoT-Based Smart Maldive Fish Processing System, was developed as an integrated Artificial Intelligence (AI) and Internet of Things (IoT) solution for different stages of Maldive fish production. The complete system combines four main research components: AI-assisted raw fish quality assessment, optimization of boiling and salt distribution, IoT-based drying environment monitoring, and multi-sensor final dried-fish quality evaluation. These components are connected through a common processing workflow and a centralized monitoring environment, allowing information generated at different production stages to contribute to a more structured and digitally supported processing process.

Within the complete workflow, the drying stage is particularly sensitive to environmental conditions. Variations in relative humidity, temperature, and airflow can influence the rate and uniformity of moisture removal from the processed fish. The overall research therefore includes a dedicated IoT-based monitoring component to continuously observe the drying microclimate and provide timely information when undesirable environmental conditions occur. This environmental monitoring function contributes to the wider goal of reducing dependence on manual observation and supporting more consistent Maldive fish processing

1.2 Individual component context

The individual research component presented in this thesis is titled Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization**.** The component focuses specifically on monitoring the environmental conditions that influence the drying stage of Maldive fish processing. The main monitored parameters are temperature, relative humidity, airflow, and atmospheric gases associated with changes in the drying environment. These parameters were selected because drying is governed by the interaction between heat, moisture transfer, air movement, and surrounding atmospheric conditions.

The developed component is centred on an ESP32-based IoT edge node connected to environmental sensors. A DHT22 sensor is used to measure dry-bulb temperature and relative humidity, an airflow sensing device is used to measure air velocity, and an MQ-135 metal-oxide semiconductor gas sensor is used to observe ammonia and amine-related gas build-up within the drying environment. The sensor node periodically collects environmental measurements and transmits the processed telemetry through MQTT over Wi-Fi for remote monitoring.

In addition to simple data collection, the component includes threshold-based environmental evaluation. The edge firmware checks combinations of environmental conditions and identifies situations that may negatively affect drying. For example, the implemented decision rule identifies a critical condition when relative humidity exceeds 75% while airflow falls below 0.5 m/s for more than 15 minutes. Under such conditions, the system generates an alert to support timely corrective action. Therefore, the component functions not only as a monitoring device but also as an early-warning mechanism within the wider smart Maldive fish processing system.

1.3 Problem statement

Traditional Maldive fish drying is commonly carried out in open yards, solar drying environments, or other processing spaces where environmental conditions are difficult to maintain at a constant level. Changes in weather, particularly increases in humidity during rainfall and reductions in airflow, can slow the drying process and create unsuitable conditions around the product. When such changes are identified only through manual observation, operators may not respond quickly enough to prevent inefficient drying or deterioration of product quality. The research paper identifies unmonitored humidity increases and stagnant boundary-layer airflow as important weaknesses within conventional drying practices.

The problem is not limited to the absence of temperature monitoring. Drying conditions are affected by several interacting environmental variables, meaning that a single-parameter monitoring approach may not provide sufficient information about the state of the processing environment. The PP1 research design therefore identified the need to monitor temperature, humidity, ammonia-related gases, and airflow as a combined environmental profile rather than as isolated measurements. It also identified the need for real-time monitoring, early warnings, and historical environmental information to support operators during the drying process.

Accordingly, the central problem addressed by this research is the lack of a continuous, integrated, and responsive environmental monitoring mechanism specifically designed for Maldive fish drying. Without such a system, sudden environmental changes may remain unnoticed until they have already affected drying performance. This research addresses that problem by developing a multi-parameter IoT monitoring subsystem capable of continuously collecting environmental data, transmitting the readings in real time, and generating alerts when predefined unsafe drying conditions persist.

1.4 Research gap

Previous research has explored IoT-based monitoring, computer vision, electronic sensing, and automated drying technologies in food processing and fish quality assessment. However, many existing studies focus on individual technologies or isolated stages of the processing lifecycle. Research discussed in the project paper includes computer vision approaches for fish freshness classification, electronic-nose or metal-oxide gas sensing for spoilage assessment, and automated drying systems. These approaches are useful within their intended applications, but they are generally not integrated into a complete Maldive fish processing workflow.

A further gap exists in environmental monitoring during the drying of processed fish. Much of the existing literature concentrates on raw fish freshness, cold-chain storage, or general agricultural monitoring. Comparatively little research addresses the specific environmental conditions experienced during the drying of boiled, salted, and cured Maldive fish, where the physical and chemical characteristics of the product change throughout a multi-day processing period.

The individual research component addresses this gap by combining multiple environmental parameters within a single IoT monitoring framework. Rather than monitoring temperature and humidity alone, the developed approach integrates temperature, relative humidity, airflow, and atmospheric gas sensing with real-time MQTT telemetry and threshold-based alert generation. The contribution is therefore not simply the use of IoT technology, but its application as a multi-parameter, real-time environmental monitoring subsystem specifically within Maldive fish drying.

1.5 Aim and objectives

Aim

The aim of this research is to design, implement, and evaluate an Intelligent IoT-Based Multi-Parameter Environmental Monitoring System capable of continuously monitoring critical environmental conditions during Maldive fish drying and providing timely information and alerts to support more stable and efficient processing conditions.

Objectives

The specific objectives of the research are:

1. To identify the key environmental parameters that influence Maldive fish drying, with particular attention to temperature, relative humidity, airflow, and atmospheric gas conditions.
2. To design and implement an ESP32-based multi-sensor IoT monitoring node for continuous environmental data acquisition during fish drying.
3. To collect, process, and transmit environmental sensor data through an IoT communication mechanism for real-time monitoring.
4. To implement a threshold-based environmental evaluation and alert mechanism capable of identifying persistent unsafe drying conditions.
5. To evaluate the technical performance of the developed monitoring component using recorded telemetry, data transmission reliability, alert responsiveness, and observed drying behaviour.

These objectives are consistent with the original project proposal, which identified real-time environmental monitoring, cloud-based data handling, threshold-based alerts, and performance evaluation as the main goals of the individual component.

1.6 Research questions or hypotheses

This study was guided by the following research questions:

RQ1: How can multiple environmental parameters, including temperature, relative humidity, airflow, and atmospheric gas conditions, be continuously monitored within a Maldive fish drying environment using an IoT-based sensing system?

RQ2: How effectively can an ESP32-based IoT monitoring node collect, process, and transmit environmental telemetry during a continuous fish drying cycle?

RQ3: Can a threshold-based monitoring approach provide timely alerts when environmental conditions move outside the defined drying envelope?

RQ4: How does continuous environmental monitoring support improved visibility and stability of the Maldive fish drying process compared with unmonitored ambient drying conditions?

The study was primarily implementation and evaluation oriented; therefore, these research questions were considered more appropriate than formulating purely statistical hypotheses. The questions directly connect the research problem, developed technical artefact, and experimental evaluation performed for the individual component.

1.7 Scope limitations and assumptions

The scope of this research is limited to the environmental monitoring of the Maldive fish drying stage. The component focuses on acquiring and evaluating temperature, relative humidity, airflow, and atmospheric gas measurements using an ESP32-based sensor node. The work includes sensor integration, periodic data acquisition, short-term filtering of telemetry, MQTT-based communication, environmental threshold evaluation, and alert generation. Integration with the wider smart Maldive fish processing architecture is also considered where environmental monitoring information is transferred to the common monitoring workflow.

The component does not perform the AI-based classification of raw fish, optimization of boiling and salt distribution, or final dried-fish image classification. These functions belong to the other individual research components within the group project. Consequently, performance measures such as raw fish classification accuracy or final product grading accuracy are not treated as results of this individual research.

The experimental system also has technical limitations. In particular, the MQ-135 metal-oxide gas sensor is sensitive to variations in ambient humidity. Significant changes in relative humidity may therefore influence sensor behaviour and can require baseline recalibration. This limitation is recognized in the overall research paper and affects the interpretation of gas measurements under rapidly changing environmental conditions.

The research further assumes that the monitored environmental variables provide meaningful indicators of the drying microclimate and that the sensor node is positioned in a location that reasonably represents the environmental conditions experienced by the drying fish. The system is intended to support operational awareness and early intervention; it is not presented as a replacement for laboratory-based microbiological or chemical analysis.

1.8 Individual contribution

The individual contribution of this research is the design, implementation, integration, and evaluation of the environmental monitoring subsystem within the wider Smart Maldive Fish Processing System. The principal technical contribution is an ESP32-based multi-sensor IoT node capable of observing multiple environmental conditions associated with the fish drying process. Unlike a simple temperature-monitoring device, the developed component brings together temperature, relative humidity, airflow, and atmospheric gas sensing to provide a broader representation of the drying microclimate.

The component also contributes a practical telemetry and decision-support workflow. Environmental measurements were sampled every 30 seconds, and the initial prototype generated 20,160 raw readings over seven days. A one-minute sliding-window median filtering process produced 10,080 validated telemetry packets for operational monitoring and MQTT transmission.

The developed firmware additionally evaluates combined environmental conditions rather than relying only on individual sensor values. A persistent high-humidity and low-airflow condition activates the alert mechanism, allowing operators to respond to adverse drying conditions. Experimental evaluation demonstrated a 99.6% MQTT packet delivery rate over a continuous seven-day drying period. During an observed rainfall event, the monitoring node detected the environmental disturbance and triggered the alert in 2.1 seconds.

Therefore, the individual contribution extends across hardware sensing, IoT communication, environmental data processing, real-time threshold evaluation, alerting, and experimental validation. This component provides the environmental intelligence required during the drying stage and forms the IoT monitoring layer between the upstream boiling and salting process and the downstream final product quality assessment stages of the overall research system.

1.9 Thesis structure

This thesis is organised into nine main chapters.

Chapter 1 introduces the overall Smart Maldive Fish Processing System and establishes the context of the individual IoT-based environmental monitoring component. It presents the problem statement, research gap, research aim and objectives, research questions, scope and limitations, and the individual contribution.

Chapter 2 reviews the literature and related work relevant to IoT-based environmental monitoring, fish drying, environmental sensing, multi-sensor systems, IoT communication, and existing approaches to food-processing monitoring. The chapter concludes by synthesising the identified research gap and establishing the conceptual basis of the proposed component.

Chapter 3 explains the research methodology adopted for the study. It describes the research design, study materials, experimental procedure, evaluation measures, data analysis approach, ethical and security considerations, and the research artefacts required to support reproducibility.

Chapter 4 presents the design of the environmental monitoring solution. It discusses the functional and non-functional requirements, the position of the component within the complete Smart Maldive Fish Processing System, major design decisions, detailed hardware and communication design, and relevant professional practices.

Chapter 5 describes the implementation, integration, and testing of the component. It explains the development environment, sensor and ESP32 implementation, MQTT-based communication, environmental monitoring logic, integration with the wider system, verification activities, and operational deployment.

Chapter 6 presents the research and technical results obtained during validation. This includes environmental telemetry behaviour, packet transmission performance, alert responsiveness, and observations obtained during the monitored drying cycle.

Chapter 7 discusses the meaning of the findings and compares the developed approach with related research and existing solutions. It also examines the research and product contributions, technical limitations, threats to validity, and practical implications.

Chapter 8 concludes the thesis by summarising the major findings, evaluating the achievement of the research objectives, identifying the individual research and product contributions, and presenting recommendations for future improvement and further research.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review examines previous research that is directly relevant to the development of the Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization. The review focuses on the technological and research areas required to support environmental monitoring during Maldive fish drying, including Internet of Things (IoT) systems, environmental sensing, fish-quality monitoring, gas sensing, drying-process behaviour, wireless communication, and real-time alert mechanisms.

The review gives particular attention to studies that explain how environmental conditions influence fish quality and dehydration. Existing research has explored automated fish freshness assessment, image-based inspection, electronic gas sensing, and drying behaviour using different technological approaches [1], [2], [7]–[10], [12], [15], [16]. These studies provide an important foundation for understanding how digital monitoring technologies can be applied to food-processing environments.

Research on automated food-quality assessment has increasingly moved away from fully manual inspection toward computer vision, machine learning, and sensor-based monitoring [1], [2]. Fish-specific studies have examined freshness classification using conventional machine-learning methods [7], multimodal freshness datasets and gas sensing [8], [9], hyperspectral imaging [10], and deep-learning-based classification [12]. Research has also investigated dried-fish quality assessment and moisture-loss behaviour during marine-product drying [15], [16].

Although the wider Smart Maldive Fish Processing System includes multiple AI- and IoT-based components, this review is intentionally focused on the environmental monitoring requirements of the drying stage. Therefore, the main emphasis is placed on temperature, relative humidity, airflow, atmospheric gas sensing, edge processing, and real-time telemetry. These areas directly correspond to the sensing and monitoring architecture implemented in the individual research component [17].

The purpose of this review is therefore to identify existing approaches, examine their limitations, and establish the research gap that justifies the development of a dedicated multi-parameter environmental monitoring system for Maldive fish drying.

2.2 Domain and theoretical foundations

The Internet of Things forms the main technological foundation of the proposed monitoring system. IoT-based systems combine physical sensors, embedded processing devices, communication networks, and digital monitoring applications to enable continuous observation of real-world environments. In a fish-processing context, this approach provides an opportunity to replace occasional manual observation with continuous environmental measurements and real-time information.

Drying is fundamentally influenced by the movement of heat and moisture between the fish and the surrounding environment. Research on marine-product drying shows that moisture removal and quality degradation are closely related to processing conditions and the surrounding atmosphere [16]. This makes temperature, relative humidity, and airflow particularly important variables during dehydration.

Temperature influences the rate of heat transfer and evaporation from the fish surface. Relative humidity affects the moisture gradient between the fish and the surrounding air. When ambient humidity becomes high, the surrounding air has a lower capacity to accept additional moisture, which can slow the drying process. Airflow is also important because moving air helps remove moisture from the boundary layer surrounding the fish. Therefore, reduced airflow may contribute to slower and less uniform dehydration [16].

Environmental gas sensing provides an additional layer of information. Previous studies have investigated electronic-nose systems and metal-oxide semiconductor sensors for detecting compounds associated with fish spoilage and deterioration [8], [9]. Volatile compounds such as ammonia and related amines can provide evidence of biochemical changes that may not be immediately visible through visual inspection alone. This makes gas sensing relevant as a supplementary environmental indicator within fish-processing environments.

The individual system developed in this research therefore combines temperature, relative humidity, airflow, and atmospheric gas monitoring through an ESP32-based IoT node [17]. Rather than treating each environmental condition independently, the design recognises that drying quality is affected by the combined state of the surrounding microclimate.

Another important theoretical concept is edge-based decision making. In conventional IoT monitoring, sensors may simply transmit readings to a remote system for later interpretation. In the developed component, part of the environmental evaluation is performed directly on the ESP32. The implemented firmware identifies an unsafe condition when relative humidity remains above 75% while airflow falls below 0.5 m/s for more than 15 minutes [17]. This approach allows abnormal environmental situations to be detected close to the sensing source and supports faster operational response.

2.3 Related research

Automated fish-quality assessment has received increasing attention in recent years. Earlier approaches generally relied on manually selected colour, texture, or shape features combined with traditional machine-learning algorithms. Yudhana *et al.* investigated fish freshness identification using k-nearest neighbours and Naïve Bayes classifiers, demonstrating the potential of machine learning to reduce dependence on subjective manual assessment [7].

Deep-learning methods later improved automated feature extraction by allowing models to learn visual characteristics directly from images. Deep learning has been applied widely across agriculture and food-related applications [3]. EfficientNet [4], MobileNetV3 [13], and residual networks [14] are examples of architectures that have contributed to improvements in image classification and feature extraction while balancing model complexity and performance.

In fish-quality applications, Kashani Zadeh *et al.* investigated hyperspectral imaging combined with deep learning for rapid freshness assessment [10]. Islam *et al.* similarly explored computer-vision-based fish freshness classification using deep-learning techniques [12]. These studies demonstrate the value of advanced visual inspection,

although they primarily focus on the product itself rather than the environmental conditions surrounding the drying process.

Gas sensing has also been investigated as a complementary approach to fish-quality assessment. Prasetyo *et al.* introduced the DaFiF dataset to support fish freshness research [8] and examined freshness classification during ice storage [9]. This work highlights the potential value of volatile-gas information when used alongside other forms of quality assessment.

Research related specifically to dried fish is more limited. Uddin *et al.* studied dried-fish quality using digital image-processing techniques [15], while Taheri and Lee investigated moisture-loss kinetics and quality degradation during marine-product drying [16]. These studies are particularly relevant because they demonstrate that both the physical state of the product and the environmental conditions during drying influence final product quality.

However, existing studies largely focus on isolated stages such as raw fish freshness, refrigerated storage, image-based inspection, or mathematical drying models. Comparatively less attention has been given to continuous, multi-parameter environmental monitoring during the drying of processed Maldive fish. The wider group research also identifies this fragmentation, noting that existing technologies often remain separated across raw fish inspection, process optimisation, environmental monitoring, and final product assessment [17].

2.4 Existing systems products and techniques

Existing automated food and fish-monitoring solutions use a range of technologies, including computer vision, embedded sensing, electronic noses, IoT communication, and cloud-based data visualisation. Computer-vision systems have become increasingly important in food-quality evaluation because they provide objective and repeatable assessment of visible characteristics [1], [2], [11].

Earlier computer-vision approaches commonly relied on handcrafted features, while more recent systems use deep neural networks to automatically learn more complex visual representations [3]. YOLO-based architectures are widely used for real-time

detection because they combine localisation and classification within a single processing pipeline [5], [6]. EfficientNet, MobileNetV3, and ResNet architectures have also demonstrated strong performance in visual classification tasks [4], [13], [14].

However, image-based methods alone do not provide continuous information about the drying environment. Environmental monitoring requires embedded sensing systems that can repeatedly measure physical conditions and transmit the results over time. In many existing IoT applications, temperature and humidity are the main variables monitored. While these measurements are useful, they may not fully represent the conditions affecting fish drying.

The current research extends this approach by including airflow and atmospheric gas sensing alongside temperature and humidity [17]. This allows the system to observe several interacting environmental factors rather than relying on a single environmental measurement.

The communication approach is also important in an IoT monitoring system. In the implemented component, processed environmental telemetry is transmitted using MQTT over Wi-Fi [17]. This lightweight communication approach is suitable for frequent sensor updates from a resource-constrained embedded device.

Another difference between monitoring systems lies in how abnormal conditions are identified. Some systems simply display sensor readings and rely on operators to decide whether intervention is needed. The current system adds local threshold-based evaluation so that persistent combinations of high humidity and low airflow can be automatically identified [17].

Despite these advantages, sensor-based systems also have limitations. The MQ-135 metal-oxide semiconductor sensor used in the project can be affected by changes in ambient humidity. Significant humidity variation may influence the sensor response and may require recalibration [17]. This limitation must therefore be considered when interpreting gas measurements in a changing drying environment.

2.5 Gap synthesis and conceptual framework

The reviewed literature demonstrates that the main technologies required for intelligent fish-processing systems already exist individually. Computer vision has been applied successfully to food and fish-quality inspection [1], [2], [7], [10], [12]. Gas sensing has been used to support fish freshness assessment [8], [9]. Image-processing methods have been applied to dried-fish quality evaluation [15], while drying models have demonstrated the importance of moisture-loss behaviour and environmental conditions [16].

However, these approaches reveal several important gaps.

First, much of the existing research focuses on raw fish freshness, storage, or final product assessment, rather than environmental conditions during the intermediate drying stage. This is particularly important for Maldive fish because the product undergoes a multi-stage process that includes boiling, salting, dehydration, and storage [17].

Second, many environmental monitoring approaches concentrate mainly on temperature and humidity. Fish drying is influenced by a broader combination of environmental factors, particularly airflow and atmospheric gas conditions. Therefore, a more comprehensive system requires simultaneous monitoring of multiple variables.

Third, many monitoring systems operate passively by collecting and displaying data. This still requires an operator to continuously observe the system and identify abnormal conditions manually. A more practical approach is to combine real-time sensing with local evaluation and automatic alert generation.

Based on these gaps, this research proposes a conceptual framework in which environmental measurements move through a sequence of sensing, processing, communication, and response stages:

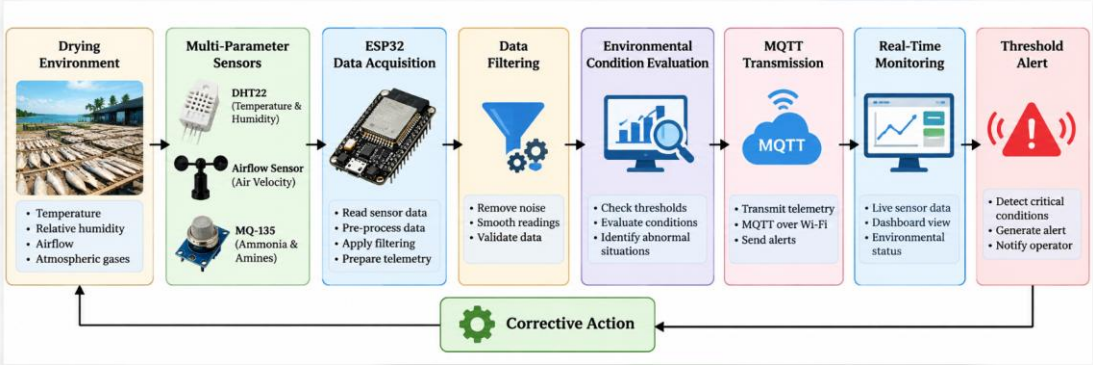


Figure 1:Conceptual Framework of the Intelligent IoT-Based Multi-Parameter Environmental Monitoring System

In this framework, the drying environment represents the physical process being observed. Environmental sensors collect temperature, relative humidity, airflow, and atmospheric gas information. The ESP32 then performs local acquisition and filtering before evaluating the measurements against the defined environmental conditions. Validated telemetry is transmitted through MQTT for real-time monitoring, while unsafe conditions trigger alerts that support corrective action [17].

This conceptual framework addresses the identified research gap by combining multi-parameter sensing, edge processing, real-time IoT communication, and deterministic alerting within a single environmental monitoring subsystem. It also provides the foundation for the methodology, solution design, implementation, and experimental evaluation presented in the following chapters.

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopted a Design Science Research (DSR**)** approach because the study focused on designing, developing, and evaluating a practical technological solution for a real-world problem in Maldive fish processing. The problem addressed was the lack of continuous and integrated environmental monitoring during the drying stage, where changes in temperature, relative humidity, airflow, and atmospheric gas conditions can affect drying stability and product quality.

The research was organized around four main activities: problem identification, artefact design and development, experimental deployment, and technical evaluation. During the first stage, environmental factors relevant to the drying process were identified from the project requirements and related literature. These requirements were then translated into the design of an ESP32-based multi-parameter monitoring node capable of collecting environmental measurements and transmitting them through an IoT communication mechanism.

The second stage involved implementation of the monitoring artefact. Temperature, relative humidity, airflow, and atmospheric gas sensing were integrated into the ESP32-based node. Local processing was used to prepare sensor readings before transmission, while deterministic threshold logic was implemented to identify environmental conditions requiring attention [17].

The third stage involved deployment of the developed component within the drying stage of the wider Smart Maldive Fish Processing System. Environmental telemetry was collected continuously during a seven-day dehydration period. The final stage involved evaluating the developed artefact using telemetry continuity, MQTT transmission reliability, threshold response behaviour, and observed drying performance [17].

This methodology was suitable because it combined system development with empirical evaluation, allowing the study to assess both the technical operation and practical relevance of the developed IoT monitoring component.

3.2 Data participants or study materials

This individual study did not use human participants as the primary source of research data. Instead, the research was based on environmental telemetry and technical system outputs generated by the developed IoT monitoring subsystem. Therefore, participant-based demographic or questionnaire data were not required.

The main study materials consisted of an ESP32 microcontroller, a DHT22 temperature and humidity sensor, a hot-wire airflow sensor, and an MQ-135 metal-oxide semiconductor gas sensor. The DHT22 was used to monitor dry-bulb temperature and relative humidity, while the airflow sensor measured air velocity within the drying environment. The MQ-135 was used to observe atmospheric ammonia and amine-related gas build-up.

The primary dataset consisted of environmental readings obtained from these sensors during the monitored drying process. The data categories included temperature, relative humidity, airflow velocity, atmospheric gas readings, telemetry transmission records, threshold events, and alert-generation events.

Environmental measurements were sampled every 30 seconds. During the initial seven-day prototype period, the system produced 20,160 raw readings. These readings were subsequently processed using a one-minute sliding-window median filtering approach, resulting in 10,080 validated telemetry packets, with one processed packet produced for each minute of operation.

Additional study materials included the ESP32 firmware, threshold conditions, MQTT communication workflow, system architecture, and monitoring outputs produced during the experimental period.

3.3 Experimental or study procedure

The experimental procedure was carried out in a sequence of stages beginning with sensor integration and ending with system-level evaluation.

First, the environmental sensing devices were connected to the ESP32 microcontroller. The DHT22, airflow sensor, and MQ-135 were configured to provide the environmental measurements required by the study. The ESP32 functioned as the central acquisition and processing device, collecting readings from the sensors at 30-second intervals.

The next stage involved preprocessing the environmental telemetry. Because short-term sensor fluctuations could reduce the stability of the monitoring data, a one-minute sliding-window median filtering process was applied. This produced a validated telemetry packet every minute and reduced the influence of temporary spikes or noise.

The processed telemetry was then transmitted through MQTT over Wi-Fi. The ESP32 published the validated environmental packets to an MQTT broker, enabling the readings to be made available to the wider monitoring environment.

At the same time, environmental conditions were evaluated locally through threshold-based logic. The implemented rule identified an unsafe drying condition when relative humidity exceeded 75% while airflow remained below 0.5 m/s for more than 15 minutes. When this combination persisted, the system generated an alert to support corrective action.

The monitoring node was then operated continuously during a seven-day dehydration cycle. During this period, environmental data, communication behaviour, and threshold events were recorded. Particular attention was given to changes in environmental conditions and the response of the system when adverse conditions occurred.

The experimental process can therefore be summarized as:

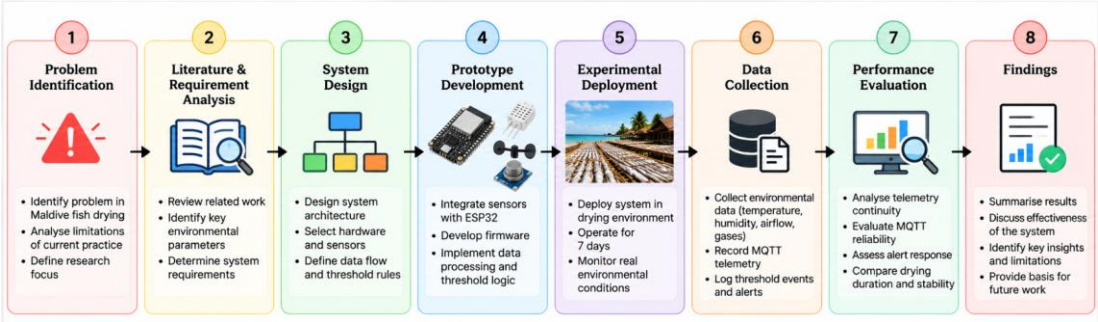


Figure 2:Research Methodology Process of the IoT-Based Environmental Monitoring Study

3.4 Measures and evaluation metrics

The evaluation measures were selected to determine whether the developed component could operate continuously, transmit reliable environmental information, and respond appropriately to adverse drying conditions.

The first evaluation measure was telemetry continuity. This considered whether environmental measurements could be collected and processed consistently throughout the monitoring period. The number of validated telemetry packets generated during the experiment was used to assess continuous system operation.

The second measure was data transmission reliability. This was evaluated using MQTT packet delivery performance. The proportion of successfully delivered telemetry packets provided an indication of the stability of the wireless communication process.

The third measure was alert response behaviour. The system was evaluated according to its ability to identify an unsafe environmental condition and generate a corresponding alert. The time taken to produce an alert following the detection of an abnormal condition was considered as part of the technical evaluation.

The fourth measure concerned environmental response and process stability. Changes in relative humidity and airflow were examined during environmental disturbances, and the behaviour of the monitoring component was analysed in relation to the corrective actions taken.

The final evaluation measure considered drying duration. The monitored dehydration period was compared with the reported unmonitored ambient drying condition to examine whether continuous environmental awareness and timely intervention were associated with more stable drying behaviour.

These metrics were selected because they directly reflected the objectives of the individual component rather than the AI-classification performance measures used by other components of the wider project.

3.5 Data analysis

The environmental data were analysed using preprocessing, descriptive analysis, event-based analysis, and comparative evaluation.

The first stage involved processing the raw sensor readings. Since measurements were generated every 30 seconds, a one-minute sliding-window median filter was applied to reduce short-term fluctuations. Median filtering was used to produce a more stable telemetry stream without removing meaningful environmental changes.

The validated environmental data were then examined over time to identify changes in temperature, relative humidity, airflow, and atmospheric gas conditions. Particular attention was given to periods where environmental conditions moved toward or beyond the defined drying thresholds.

Event-based analysis was used to examine the behaviour of the system during abnormal environmental conditions. When the threshold combination was reached, the associated sensor readings, alert response, and subsequent environmental behaviour were considered together.

Communication performance was analysed using MQTT telemetry records. The number of validated packets generated and successfully delivered was used to determine the reliability of the communication layer.

Finally, comparative analysis was used to examine the monitored dehydration duration in relation to the reported unmonitored ambient drying duration. This comparison was used to interpret whether continuous monitoring and corrective intervention were associated with improved drying stability.

The analysis remained primarily descriptive and technical because the objective of the individual study was to evaluate the operation of an IoT artefact rather than to establish population-level statistical relationships.

3.6 Ethics privacy security and responsible AI

This individual research component did not involve direct experimentation with human participants. Therefore, the main ethical considerations were related to the safe deployment of the monitoring equipment, responsible use of collected data, and appropriate interpretation of the system outputs.

The collected data consisted primarily of environmental measurements and technical system records. No personal, demographic, or sensitive participant information was required. Environmental telemetry and system logs were used only for research, technical validation, and evaluation of the developed component.

Safe deployment was also an important consideration. Sensors, the ESP32 node, and associated equipment were required to operate without unnecessarily interfering with the fish-processing environment or creating risks for people working around the system.

Security was relevant because environmental telemetry was transmitted wirelessly. Controlled system access and reliable communication were therefore considered important aspects of the prototype architecture. However, the current work remained a research prototype rather than a fully hardened commercial IoT deployment.

The individual environmental monitoring component did not use an AI model to make autonomous environmental decisions. Instead, it used transparent and deterministic threshold-based logic. This meant that the reason for each alert could be traced directly to the measured environmental conditions. AI-based functions used elsewhere in the wider Smart Maldive Fish Processing System were outside the decision-making scope of this individual component.

Generative AI tools used during preparation of the thesis were limited to support for language refinement, structure, and academic presentation. Technical claims, implementation details, and quantitative results were verified against the project

proposal, presentations, research paper, and available implementation evidence before inclusion in the final thesis.

3.7 Reproducibility and research artefacts

Reproducibility was considered throughout the research by documenting the main hardware configuration, monitored parameters, sampling strategy, filtering method, communication approach, threshold logic, and evaluation procedure.

The principal research artefacts generated through this study include the ESP32-based environmental monitoring node, sensor integration architecture, firmware logic, threshold conditions, MQTT telemetry workflow, environmental monitoring records, and experimental validation outputs.

Important operational settings were also documented. Environmental readings were sampled every 30 seconds and processed into one-minute validated telemetry packets using median filtering. The communication layer used MQTT over Wi-Fi, while the environmental alert rule considered relative humidity above 75% together with airflow below 0.5 m/s for more than 15 minutes.

The recording of these details provides a clear basis for reproducing the experimental configuration or extending the system in future work. However, exact replication may still be influenced by sensor calibration, hardware placement, local weather conditions, drying-environment characteristics, and network conditions.

The research artefacts documented in this chapter also establish traceability across the remainder of the thesis. The methodological decisions described here form the basis for the product architecture presented in Chapter 4, the implementation and testing described in Chapter 5, and the experimental results evaluated in Chapter 6.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The design of the Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization was guided by the need to continuously observe environmental conditions during the Maldive fish drying process. The system requirements were derived from the research problem, literature review, project proposal, presentation materials, and the final research paper. The main requirement was to monitor multiple environmental parameters simultaneously so that the drying environment could be assessed more comprehensively than through temperature or humidity measurements alone.

The principal functional requirements included continuous monitoring of temperature, relative humidity, airflow, and atmospheric gas conditions. The system also needed to collect sensor readings automatically, preprocess the measurements, transmit validated telemetry in real time, and generate alerts when predefined unsafe environmental conditions persisted. These requirements were consistent with the wider project objective of improving environmental visibility and reducing dependence on manual observation during the drying stage.

The system was also required to support continuous operation over an extended drying period. For this reason, regular data acquisition, filtering, and reliable wireless communication were considered essential. Environmental readings were sampled every 30 seconds and processed into one-minute validated telemetry packets before transmission.

Several non-functional requirements were also considered during the design. The component needed to be modular, maintainable, and capable of integrating with the wider Smart Maldive Fish Processing System. The architecture also needed to support future extension, allowing sensors, threshold rules, or communication functions to be modified without redesigning the entire subsystem.

4.2 Component architecture within the complete solution

The environmental monitoring component operates as one part of the wider IoT-Based Smart Maldive Fish Processing System. The complete solution follows the Maldive fish production lifecycle from raw fish assessment through boiling and salting, drying, and final quality evaluation.

Within this overall architecture, the individual component is positioned specifically at the drying and microclimate monitoring stage. Once the boiling and salting stage is completed, the processed fish enters the drying environment, where the IoT monitoring node continuously observes environmental conditions. This allows drying-stage information to contribute to the wider digital processing workflow.

The component architecture is organised into four main layers. The first is the sensing layer, which captures temperature, relative humidity, airflow, and atmospheric gas measurements. The second is the edge-processing layer, where the ESP32 acquires the sensor readings, applies preprocessing, and evaluates threshold conditions. The third is the communication layer, where processed telemetry is transmitted through MQTT over Wi-Fi. The fourth is the monitoring and response layer, where environmental information can be viewed and alerts can support corrective action.

The PP2 system architecture identified ESP32 sensor communication, sensor data transmission, AI model outputs, and a central dashboard as important integration points across the complete system. Within this structure, the environmental monitoring component contributes drying-stage telemetry and alert information while remaining functionally separate from the raw fish assessment, boiling and salting optimisation, and final quality assessment modules.

4.3 Design alternatives and decisions

Several design alternatives were considered before finalising the environmental monitoring architecture. One important decision concerned whether the system should monitor only temperature and humidity or adopt a broader multi-parameter sensing approach.

A simpler temperature-and-humidity-only design would have reduced hardware complexity. However, this approach would not have captured important information about airflow or atmospheric gas conditions. Since drying behaviour is influenced by several interacting environmental variables, a multi-parameter monitoring approach was considered more appropriate.

Another design decision concerned the embedded processing platform. The ESP32 microcontroller was selected because it could support multiple sensor connections, local data processing, threshold evaluation, and Wi-Fi communication within a compact embedded device.

The communication method was also considered during the design stage. Although the original project planning considered common IoT communication approaches such as HTTP and MQTT, the implemented system used MQTT for environmental telemetry transmission. MQTT was suitable for the prototype because the system generated frequent, relatively small sensor messages that needed to be transmitted efficiently.

A further design choice involved data preprocessing. Transmitting every raw reading directly could introduce unnecessary short-term fluctuations into the monitoring stream. Therefore, the system used a one-minute sliding-window median filter to process the 30-second sensor readings before operational transmission. This reduced temporary noise while preserving meaningful environmental changes.

The system also performed threshold evaluation directly at the edge rather than relying entirely on remote processing. This decision allowed adverse conditions to be identified locally and supported faster alert generation.

4.4 Detailed design

The detailed design of the component consists of six connected functional stages: environmental sensing, ESP32 data acquisition, median filtering, threshold evaluation, MQTT transmission, and real-time monitoring with alert response.

The environmental sensing stage uses three main sensing devices. The DHT22 measures dry-bulb temperature and relative humidity. A hot-wire airflow sensor measures air velocity within the drying environment. The MQ-135 metal-oxide semiconductor gas sensor is used to observe atmospheric ammonia and amine-related gas build-up.

The ESP32 acts as the central acquisition and processing unit. It receives sensor measurements at 30-second intervals and prepares the readings for further processing. During the prototype evaluation, this configuration generated 20,160 raw readings over the seven-day monitoring period.

A one-minute sliding-window median filter is then applied to the raw measurements. This filtering stage reduces the influence of temporary fluctuations and produces one validated environmental telemetry packet per minute. A total of 10,080 validated packets were generated during the seven-day monitoring period [17].

After preprocessing, the ESP32 evaluates the environmental conditions using deterministic threshold logic. The implemented decision rule identifies an unsafe condition when relative humidity remains above 75% while airflow remains below 0.5 m/s for more than 15 minutes.

The validated telemetry is then transmitted through MQTT over 2.4 GHz Wi-Fi to the wider monitoring environment. When unsafe conditions are detected, the system generates an alert to support timely corrective action.

The detailed operational flow can therefore be represented as:

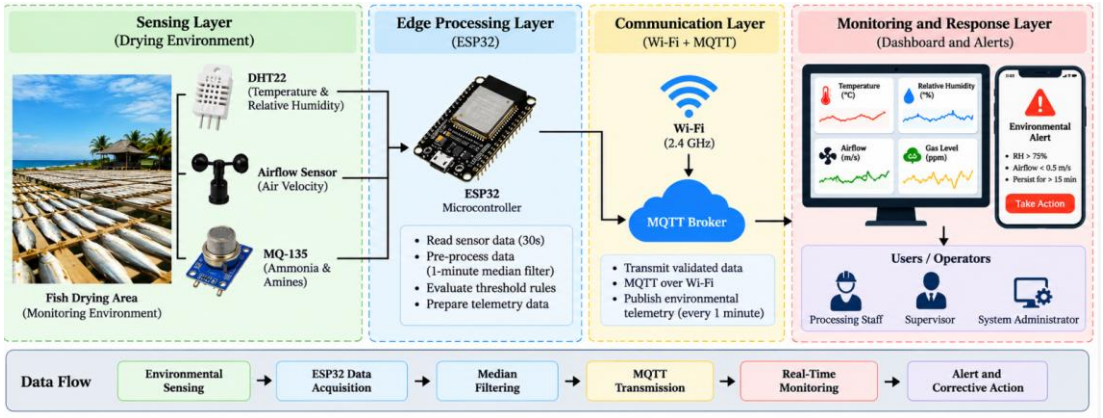


Figure 3:Detailed Architecture of the IoT-Based Multi-Parameter Environmental Monitoring Component

This modular architecture allows the sensing, processing, communication, and alerting functions to be modified independently, supporting future maintainability and system expansion.

4.5 Standards and professional practice

The development of the environmental monitoring component followed general principles of professional IoT system design, including modularity, transparency, reliability, traceability, and responsible interpretation of sensor data.

One important professional consideration was sensor reliability. The MQ-135 is a metal-oxide semiconductor sensor, and its behaviour can be influenced by changes in ambient humidity. The research paper identifies this cross-sensitivity as a technical limitation and notes that significant humidity changes may require baseline recalibration. Therefore, gas-sensor readings were treated as environmental indicators rather than laboratory-grade chemical measurements.

The system also used deterministic threshold logic rather than an opaque automated decision mechanism. This allowed the reason for each environmental alert to be understood directly from the measured conditions, improving transparency and traceability.

Data quality was supported through the use of median filtering before telemetry transmission. This reduced the influence of temporary sensor fluctuations and

produced a more stable operational data stream. Communication reliability was also evaluated during continuous operation, and the prototype achieved a 99.6% MQTT packet delivery rate during the seven-day dehydration cycle.

Finally, the architecture was designed to support future improvement. Additional sensing devices, enhanced calibration methods, stronger communication security, predictive analytics, or automated environmental control could be integrated without fundamentally redesigning the existing subsystem. This modularity supports maintainability, scalability, and continued development of the environmental monitoring solution.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization was implemented using an embedded IoT architecture designed for continuous operation in the Maldive fish drying environment. The development environment combined sensor hardware, an ESP32 microcontroller, embedded firmware, Wi-Fi communication, MQTT-based telemetry, and a monitoring interface. The technology selection was based on the need for low-cost sensing, local processing, wireless communication, and integration with the wider Smart Maldive Fish Processing System.

The ESP32 microcontroller was used as the central processing and communication device. It was responsible for acquiring environmental readings, applying data preprocessing, evaluating threshold conditions, preparing telemetry, and transmitting the processed measurements through Wi-Fi. The use of ESP32 reduced the need for additional communication hardware because Wi-Fi functionality was available within the same embedded platform.

Three main sensing devices were integrated with the ESP32. The DHT22 sensor was used to measure dry-bulb temperature and relative humidity. A hot-wire airflow sensing device measured air velocity within the drying environment. An MQ-135 metal-oxide semiconductor gas sensor was used to observe atmospheric gas build-up associated with ammonia and related amines. Together, these sensors allowed the system to monitor multiple environmental factors relevant to the fish drying process.

The communication layer used MQTT over 2.4 GHz Wi-F**i**. Processed telemetry packets were published through MQTT to the wider monitoring environment. This communication method supported frequent transmission of relatively small sensor messages and enabled the drying-stage monitoring component to participate in the complete IoT system.

The firmware included periodic sensor acquisition, local preprocessing, median filtering, environmental threshold evaluation, and telemetry preparation. Environmental measurements were sampled every 30 seconds and subsequently converted into validated one-minute telemetry packets before transmission. This combination of embedded sensing, local processing, and lightweight wireless communication formed the main technology stack used for the implemented component.

5.2 Component implementation

The component implementation began with the physical and logical integration of the environmental sensors with the ESP32. Each sensor was assigned a specific role within the monitoring system so that the combined measurements could provide a broader representation of the drying microclimate.

The DHT22 continuously supplied temperature and relative humidity measurements, while the airflow sensor captured the movement of air through the drying environment. The MQ-135 provided additional atmospheric gas information associated with ammonia and amine-related compounds. The ESP32 collected these measurements as the main embedded processing node.

Sensor values were acquired at 30-second intervals. This sampling frequency allowed the monitoring system to capture environmental changes at a sufficiently fine time scale without creating an unnecessarily large operational data stream. During the initial seven-day evaluation, the monitoring process produced 20,160 raw readings.

Short-term sensor fluctuations could reduce the clarity of the environmental information if every raw measurement were used directly. Therefore, the implementation included a one-minute sliding-window median filter. The median filtering process reduced temporary spikes and produced a more stable operational data stream. Following filtering, the seven-day raw dataset resulted in 10,080 validated telemetry packets, equivalent to one packet per minute.

The processed telemetry was then prepared for MQTT publication. The ESP32 transmitted the validated data through Wi-Fi to the MQTT broker, making the environmental readings available for real-time monitoring.

The component also included local environmental decision logic. Rather than acting only as a passive data acquisition unit, the firmware continuously evaluated combinations of environmental measurements. The implemented rule identified an unsafe condition when relative humidity remained above 75% while airflow remained below 0.5 m/s for more than 15 minutes. When this condition persisted, the system generated an alert so that corrective action could be taken.

This implementation resulted in a complete embedded monitoring workflow that combined sensor acquisition, data preprocessing, threshold evaluation, telemetry transmission, and alert generation within a single IoT node.

5.3 Integration with the other components

The environmental monitoring subsystem was integrated into the wider IoT-Based Smart Maldive Fish Processing System rather than operating as an isolated monitoring device. The overall system covers several stages of Maldive fish production, including raw fish assessment, boiling and salting optimization, drying-stage environmental monitoring, and final dried-fish quality evaluation.

Within this architecture, the individual component is positioned specifically during the drying stage. It receives no responsibility for raw fish image classification, boiling parameter optimization, salt-distribution calculations, or final dried-fish grading. Instead, its role is to provide reliable information about the environmental conditions experienced during dehydration.

The main integration output of the component is the validated environmental telemetry generated by the ESP32 node. These readings are transmitted through the IoT communication layer and made available within the wider monitoring environment. The component also contributes alert information when predefined unsafe conditions occur.

The project integration architecture identified ESP32 sensor communication, sensor data transmission, AI outputs, and the central dashboard as important connection points across the group solution. This arrangement allowed each individual component to remain modular while still contributing to the complete processing workflow.

The modular design also reduced unnecessary dependencies between the different research components. For example, changes to the environmental sensing logic could be made without modifying the raw fish assessment model or the final product quality evaluation process. This separation improved maintainability and supported clearer individual research contributions.

5.4 Verification and software testing

Verification and testing were carried out to determine whether the developed component performed its intended functions reliably under continuous operation. The main areas of evaluation were sensor acquisition, filtering behaviour, telemetry transmission, threshold detection, and alert generation.

The first stage of verification involved confirming that environmental measurements could be acquired continuously from the connected sensing devices. Temperature, relative humidity, airflow, and atmospheric gas readings were monitored throughout the test period. The successful generation of a continuous validated telemetry sequence provided evidence that the acquisition and preprocessing workflow remained operational.

The second stage examined the data filtering mechanism. Raw readings collected every 30 seconds were processed using the one-minute sliding-window median filter. The expected relationship between the raw and filtered datasets was observed, with 20,160 raw readings producing 10,080 validated one-minute telemetry packets.

The third stage focused on MQTT communication reliability. During the seven-day dehydration cycle, the system achieved a 99.6% MQTT packet delivery rate across the validated telemetry stream. This result demonstrated that the communication layer remained highly reliable during extended monitoring.

The fourth stage evaluated threshold detection under an actual environmental disturbance. On Day 3 of testing, an unexpected tropical rainfall event caused relative humidity to increase from 54% to 86% within 22 minutes, while airflow fell to 0.1 m/s. The monitoring node detected the unsafe environmental situation and generated an alert within 2.1 seconds.

Following the warning, corrective actions were taken in the drying environment, including closing the solar chamber awnings and activating auxiliary exhaust fans. These actions helped maintain chamber humidity below 68% during the disturbance.

The testing therefore validated the main operational functions of the component: continuous sensing, preprocessing, telemetry transmission, environmental condition detection, and timely alert generation.

5.5 Deployment and operation

The developed environmental monitoring subsystem was deployed within the drying stage of the wider Smart Maldive Fish Processing System and operated continuously during a seven-day dehydration cycle.

During normal operation, the sensors acquired environmental measurements every 30 seconds. The ESP32 received these readings, processed them through the one-minute median filter, evaluated the threshold conditions, and prepared validated telemetry for transmission.

The processed environmental data were then transmitted through MQTT over Wi-Fi to the monitoring environment. This allowed the drying conditions to be observed continuously rather than relying on occasional manual measurements.

The operational sequence of the deployed component can be summarized as follows:

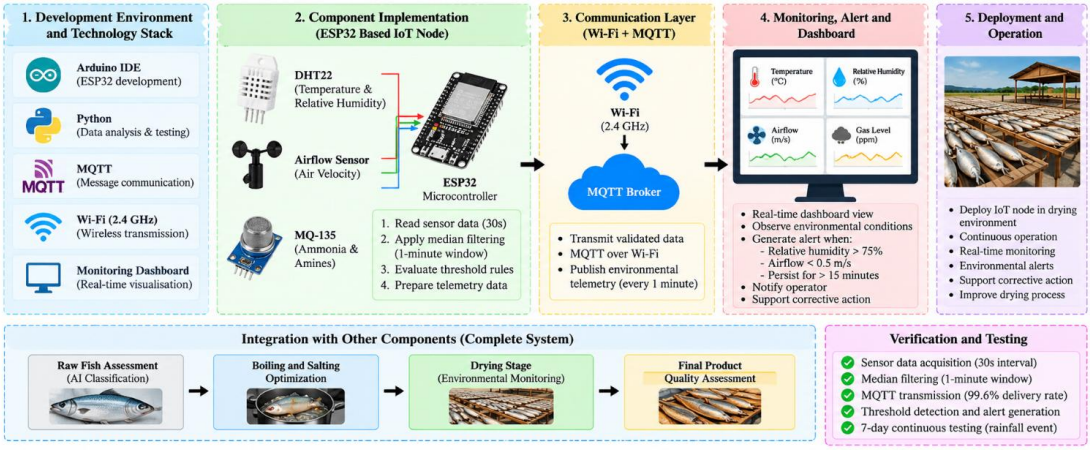


Figure 4:Implementation, Integration and Testing Architecture of the IoT-Based Environmental Monitoring Component

During the evaluation period, the component maintained a 99.6% MQTT packet delivery rate and successfully generated an alert during the rainfall-related disturbance. These results demonstrate that the system could operate continuously and respond to meaningful environmental changes during the drying process.

The monitored dehydration period was reported as 4.2 ± 0.3 days, compared with 6.5 ± 1.4 days under unmonitored ambient conditions. This difference should not be interpreted as being caused by the sensing component alone. Instead, the monitoring system provided timely environmental information that supported corrective actions, which in turn contributed to improved control of the drying environment.

The implemented solution remained a research prototype rather than a fully commercial industrial deployment. Nevertheless, the deployment demonstrated the practical feasibility of using a multi-parameter ESP32-based IoT monitoring node for continuous environmental observation and alerting within Maldive fish drying.

6 RESULTS AND VALIDATION

6.1 Research results

The experimental evaluation showed that the developed Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization was able to operate continuously during the monitored Maldive fish drying period. The evaluation focused on telemetry continuity, communication reliability, environmental disturbance detection, alert responsiveness, and the observed behaviour of the drying process.

During the seven-day dehydration cycle, the ESP32-based monitoring node generated 10,080 validated one-minute telemetry packets from the processed environmental data. These packets represented the operational data stream used for continuous monitoring of temperature, relative humidity, airflow, and atmospheric gas conditions.

A significant environmental disturbance occurred on Day 3 of testing when an unexpected rainfall event caused relative humidity to increase from 54% to 86% within 22 minutes, while airflow dropped to 0.1 m/s. This event provided a practical test of the threshold-based monitoring mechanism.

The system detected the unsafe environmental condition and generated an alert within 2.1 seconds. Following the alert, corrective actions were taken within the drying environment, including the use of protective covers and auxiliary ventilation. These actions helped maintain the chamber humidity below 68% during the disturbance.

The monitored dehydration process was reported as stabilising at 4.2 ± 0.3 days, compared with 6.5 ± 1.4 days under unmonitored ambient conditions [17]. This result indicates that continuous environmental monitoring, combined with timely intervention, was associated with more stable drying behaviour.

6.2 Product and technical results

The technical results confirmed that the developed prototype successfully performed the main functions defined during the design and implementation stages.

The IoT node was able to continuously acquire environmental readings, apply preprocessing, evaluate threshold conditions, and transmit validated telemetry through MQTT. The system achieved a 99.6% MQTT packet delivery rate during the seven-day monitoring period. This result demonstrates that the communication layer remained highly reliable during continuous operation.

The data-processing mechanism also functioned as intended. Environmental values were sampled every 30 seconds, producing 20,160 raw readings over seven days. After one-minute median filtering, the system generated 10,080 validated packets for real-time monitoring.

The threshold-based alert mechanism also performed effectively. During the rainfall-related disturbance, the system recognized the unsafe environmental condition and generated an alert within 2.1 seconds . This rapid response demonstrates the benefit of performing condition evaluation locally at the edge instead of relying entirely on manual monitoring.

Overall, the technical results validated the complete operational sequence:

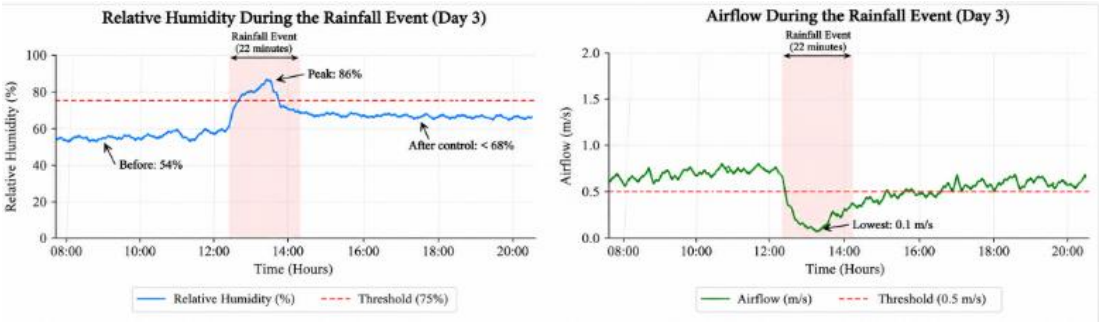


Figure 5:Rainfall event humidity + airflow graph

6.3 User or stakeholder validation

The available project evidence does not document a separate formal usability study, questionnaire, interview-based evaluation, or quantified stakeholder satisfaction assessment specifically for this individual component. Therefore, no participant numbers, satisfaction percentages, usability scores, or acceptance ratings are claimed in this thesis.

Instead, stakeholder relevance was considered through the practical requirements addressed by the system. These included the need for continuous environmental monitoring, early detection of unsuitable drying conditions, real-time visibility, and timely alert generation.

The rainfall-related event demonstrated that the system could identify an environmental disturbance and provide a warning quickly enough to support corrective action. This provides evidence of practical usefulness within the drying workflow, even though it does not constitute formal user acceptance testing.

A structured stakeholder study with fish processors, supervisors, or system operators would therefore be valuable in future work to evaluate usability, alert clarity, perceived usefulness, and long-term technology acceptance.

6.4 Integration results

The environmental monitoring component successfully operated within the wider IoT-Based Smart Maldive Fish Processing System as the subsystem responsible for the drying stage.

The main integration output was the validated environmental telemetry generated by the ESP32 node. This information was transmitted through MQTT and made available to the wider monitoring environment. The component also produced alert information when the defined environmental threshold condition was detected.

Within the overall processing workflow, the environmental monitoring subsystem remained functionally separate from the raw fish assessment, boiling and salting optimisation, and final dried-fish quality assessment components. This modular

structure allowed the drying-stage component to be developed and evaluated independently while still contributing to the broader end-to-end system.

The integration results therefore confirm that the component fulfilled its intended role as the environmental intelligence layer of the drying stage.

6.5 Summary of findings

The evaluation produced several important findings.

First, the system maintained continuous environmental monitoring over a seven-day dehydration period and generated 10,080 validated telemetry packets.

Second, the communication layer achieved a 99.6% MQTT packet delivery rate, demonstrating reliable telemetry transmission during extended operation [17].

Third, the threshold-based alert mechanism successfully responded to a real environmental disturbance. During the rainfall event, relative humidity increased from 54% to 86%, airflow decreased to 0.1 m/s, and the system generated an alert within 2.1 seconds.

Fourth, the monitored drying process was reported as stabilising at 4.2 ± 0.3 days, compared with 6.5 ± 1.4 days under unmonitored ambient conditions [17].

Finally, the results demonstrated the practical feasibility of combining multi-parameter sensing, ESP32 edge processing, median filtering, MQTT communication, and deterministic threshold-based alerting within a single IoT subsystem for Maldive fish drying.

Overall, the findings support the main research objective by showing that the developed system can provide reliable real-time environmental monitoring and timely alerts to support improved control of the fish drying environment.

7 DISCUSSION

7.1 Interpretation of findings

The findings of this research indicate that the developed Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization was able to provide continuous, structured, and responsive environmental monitoring during the Maldive fish drying process. The results demonstrate that the sensing, preprocessing, communication, monitoring, and alerting functions were able to operate together as an integrated IoT workflow over an extended drying period.

One of the most important findings was the generation of 10,080 validated one-minute telemetry packets during the seven-day dehydration cycle, together with a 99.6% MQTT packet delivery rate [17]. This result indicates that the monitoring node was able to maintain a highly consistent communication stream during continuous operation. For an IoT-based environmental monitoring system, communication reliability is essential because missing or delayed data can reduce the effectiveness of real-time monitoring and may prevent the timely identification of abnormal conditions.

The use of one-minute median filtering also contributed to the stability of the telemetry stream. Raw environmental values were initially collected every 30 seconds, but these measurements may contain temporary fluctuations caused by short-lived environmental changes or sensor noise. By applying a median filter before transmission, the system reduced the influence of short-duration spikes while retaining the underlying environmental trend. This was important because environmental monitoring during fish drying requires enough sensitivity to detect meaningful changes without reacting unnecessarily to every small fluctuation.

The rainfall-related disturbance recorded on Day 3 provided particularly strong evidence of the practical value of the developed system. During the event, relative humidity increased from 54% to 86% within 22 minutes, while airflow decreased to 0.1 m/s [17]. These measurements represented a clear shift away from the preferred drying conditions and created a useful real-world test for the threshold-based monitoring mechanism.

The system generated an alert within 2.1 seconds after identifying the unsafe environmental condition [17]. This demonstrates the advantage of edge-based evaluation, where the ESP32 processes the sensor measurements locally rather than depending entirely on a remote server or human operator to interpret the data. Local evaluation reduces the delay between environmental change and system response, which is particularly important when sudden weather conditions can rapidly affect the drying environment.

The corrective actions taken after the alert, including the use of protective covers and auxiliary ventilation, helped maintain the chamber humidity below 68% [17]. This result suggests that the developed system provided useful decision-support information rather than simply collecting data for later analysis. The monitoring node identified the adverse condition, generated a timely warning, and supported operational intervention.

The monitored dehydration process was reported as stabilising at 4.2 ± 0.3 days, compared with 6.5 ± 1.4 days under unmonitored ambient conditions [17]. This comparison indicates that the monitored process was associated with a shorter and more stable drying duration. However, this result must be interpreted carefully. The improvement cannot be attributed solely to the IoT monitoring hardware because the physical corrective actions taken after the alerts also directly influenced the drying environment. Therefore, the main value of the monitoring component is its ability to provide timely environmental awareness that enables more effective intervention.

Overall, the findings show that the system achieved more than simple environmental data collection. It functioned as a practical monitoring and early-warning subsystem capable of providing continuous visibility into the drying microclimate and supporting more informed operational decisions.

7.2 Comparison with literature baselines and existing solutions

The developed component can be compared with several categories of existing research, including automated fish freshness assessment, computer-vision-based quality inspection, electronic gas sensing, and drying-process studies.

Previous fish-quality research has concentrated heavily on the assessment of raw fish freshness. Machine-learning techniques have been used to classify freshness based on selected characteristics [7], while hyperspectral imaging and deep-learning approaches have been explored for more advanced non-destructive evaluation [10], [12]. These approaches are valuable for assessing the quality state of the fish itself, but they do not continuously observe the environmental conditions that influence the drying process after boiling and salting.

Computer-vision research has also improved automated food-quality assessment by providing repeatable inspection of visible characteristics [1], [2], [11]. Modern architectures such as EfficientNet, MobileNetV3, ResNet, and YOLO have contributed to advances in automated visual analysis [4]–[6], [13], [14]. However, visual inspection remains fundamentally different from environmental monitoring. A camera-based system may identify visible product characteristics, but it cannot independently provide continuous measurements of humidity, airflow, or atmospheric gas conditions surrounding the drying fish.

Sensor-based approaches offer a complementary perspective. Previous studies have investigated electronic-nose and gas-sensing methods for detecting volatile compounds associated with fish deterioration [8], [9]. These studies demonstrate the value of using non-visual environmental or chemical indicators. The present research extends this idea by integrating gas sensing with temperature, humidity, and airflow monitoring within one embedded IoT node.

Research on dried fish and marine-product processing also provides an important baseline. Uddin *et al.* investigated image-based quality evaluation of dried fish [15], while Taheri and Lee examined moisture-loss kinetics and quality degradation during the drying of marine products [16]. These studies reinforce the importance of controlled dehydration and demonstrate that drying behaviour is influenced by environmental and process conditions.

The present study differs from these earlier approaches in several ways. First, it focuses specifically on the drying environment rather than only on the product. Second, it uses multiple environmental parameters simultaneously, combining

temperature, relative humidity, airflow, and atmospheric gas information. Third, it integrates edge-based threshold evaluation and MQTT communication, allowing the system to provide real-time alerts rather than functioning only as a passive measurement platform [17].

Another important difference is system integration. The developed environmental monitoring component operates within the wider Smart Maldive Fish Processing System, positioned between boiling and salting optimization and final quality assessment. This creates a more connected processing workflow than many isolated monitoring or inspection solutions.

7.3 Research contribution

The primary research contribution of this study is the development and validation of a multi-parameter IoT monitoring framework specifically designed for Maldive fish drying. The contribution lies not in the individual technologies themselves, but in their coordinated application to a specific processing problem.

The system combines temperature and relative humidity sensing, airflow measurement, atmospheric gas monitoring, embedded processing, median filtering, MQTT communication, and deterministic threshold evaluation within a single architecture. This creates a practical monitoring framework that can be reproduced or adapted for similar food-drying environments.

A further research contribution is the use of combined-condition decision logic. Rather than evaluating each parameter independently, the system identifies unsafe conditions based on the interaction between relative humidity and airflow. The rule requiring relative humidity to remain above 75% while airflow remains below 0.5 m/s for more than 15 minutes provides a more context-aware interpretation of the drying environment than a single isolated threshold .

The study also contributes field-oriented evidence from a continuous seven-day monitoring period. Many prototype studies are limited to short laboratory tests, but the present evaluation included continuous operation and an actual environmental disturbance. The high packet delivery rate and successful alert response therefore

provide practical evidence that the proposed architecture can function under changing real-world conditions.

Another contribution is the separation between monitoring intelligence and physical intervention. The system itself does not automatically control the complete drying environment; instead, it detects and communicates unsafe conditions. This design makes the monitoring logic transparent and allows operators to understand the reason for each alert.

7.4 Product contribution

The developed component also makes a practical product-level contribution to the wider Smart Maldive Fish Processing System.

The prototype provides a functional environmental monitoring layer for the drying stage. It captures multiple environmental variables, processes the data locally, transmits validated telemetry, and provides alerts when the drying environment becomes unsuitable. This replaces a purely manual observation approach with continuous digital monitoring.

The component is modular by design. The sensing layer, processing logic, communication layer, and monitoring interface can be modified independently. For example, a different gas sensor could be introduced in future without changing the overall system architecture. Similarly, threshold rules or communication mechanisms could be updated without redesigning the entire monitoring node.

The product also contributes to the wider integration of the complete system. Environmental telemetry generated during drying can be viewed alongside information from the other stages of Maldive fish processing. This creates a more complete digital representation of the production workflow and supports traceability across different stages.

The prototype remains a research system rather than a fully commercial product. However, it demonstrates a clear pathway toward a more practical deployment in small-scale or semi-industrial fish-processing environments.

7.5 Limitations and threats to validity

Despite the positive findings, several limitations should be considered when interpreting the results.

The first limitation relates to the MQ-135 metal-oxide semiconductor gas sensor. This type of sensor can be influenced by ambient humidity, which means that substantial changes in relative humidity may affect the sensor response. In a drying environment, where humidity can change rapidly, this cross-sensitivity introduces uncertainty into the interpretation of gas measurements. Future versions would benefit from humidity compensation or improved calibration procedures.

The second limitation concerns the duration of the evaluation. The system was tested during a single seven-day dehydration cycle. Although this was sufficient to demonstrate continuous operation, it does not provide evidence across multiple seasons, processing locations, or weather patterns. A larger number of repeated trials would provide stronger evidence of reliability and generalisability.

A third limitation is the lack of a formal usability or stakeholder study. The current research provides technical evidence of system performance, but it does not quantify operator satisfaction, perceived usefulness, ease of use, or long-term adoption. These factors are important for practical deployment and should be evaluated separately.

Another limitation concerns the interpretation of the drying-duration result. The shorter monitored drying duration was observed together with corrective actions. Therefore, the study cannot isolate the independent effect of the monitoring system from the physical effect of ventilation and protective covers.

The threshold values also represent a potential threat to validity. The implemented humidity and airflow thresholds were tested within the current processing context, but they may not be optimal for all fish species, sizes, salting levels, drying methods, or environmental conditions. Broader experimentation would be required before applying the same rules universally.

Sensor placement may also influence the recorded environmental values. A single monitoring node represents only the local conditions around its installation point. Larger drying areas may contain spatial variation in humidity, airflow, and

temperature. This means that a single-node configuration may not capture every microclimatic difference across a large drying environment.

Finally, communication reliability may depend on local Wi-Fi availability. Although the prototype achieved a very high MQTT packet delivery rate, performance in larger or more obstructed processing environments may differ.

7.6 Implications

The findings of this research have several practical, technical, and research implications.

From a practical perspective, the study demonstrates that continuous IoT-based environmental monitoring can support traditional fish-drying operations without fundamentally changing the production method. Instead of replacing existing processing practices, the system provides additional digital visibility and timely information that can help operators make more informed decisions.

The research also highlights the value of multi-parameter monitoring. Temperature and humidity provide useful information, but they do not fully describe the drying environment. Airflow adds important information about moisture removal, while atmospheric gas measurements provide additional environmental context. Monitoring these variables together creates a more complete view of the process.

From a technical perspective, the successful use of ESP32-based edge processing demonstrates that relatively compact embedded hardware can support sensing, filtering, decision logic, and network communication within a single platform. This reduces system complexity and provides a practical foundation for future smart-processing applications.

The system architecture also provides a basis for future automation. In the current prototype, alerts support human intervention. Future systems could connect the monitoring node directly to ventilation fans, protective covers, or other actuators to automatically respond when unsafe conditions are detected.

Predictive monitoring is another possible implication. As larger environmental datasets become available, historical telemetry could be used to identify patterns that

occur before unsafe conditions develop. This could allow the system to move from reactive alerting toward predictive environmental management.

The research also has broader implications for the digitalisation of traditional Sri Lankan food-processing activities. Many traditional processing methods depend heavily on operator experience. Introducing low-cost IoT monitoring can help preserve these traditional methods while improving consistency, traceability, and process visibility.

Overall, the findings suggest that multi-parameter IoT monitoring can provide a practical bridge between traditional food-processing practices and modern digital process control.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research developed and evaluated an Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization as the drying-stage monitoring component of the wider Smart Maldive Fish Processing System. The study addressed a practical limitation in conventional Maldive fish drying, where environmental conditions are often observed manually and changes in temperature, relative humidity, airflow, and atmospheric gas conditions may not be identified quickly enough to support timely intervention.

The developed solution used an ESP32-based IoT edge node integrated with environmental sensors to continuously observe the drying environment. The DHT22 was used to measure temperature and relative humidity, an airflow sensor was used to measure air velocity, and the MQ-135 sensor was used to observe ammonia and amine-related atmospheric gas conditions. Environmental readings were sampled at 30-second intervals, processed using a one-minute median-filtering approach, and transmitted through MQTT over Wi-Fi for real-time monitoring.

The system also incorporated deterministic threshold-based decision logic. A critical drying condition was identified when relative humidity remained above 75% while airflow remained below 0.5 m/s for more than 15 minutes. This enabled the component to operate not only as a sensing platform, but also as an early-warning mechanism capable of supporting corrective action.

The experimental evaluation demonstrated continuous operation during a seven-day dehydration cycle. The system generated 10,080 validated one-minute telemetry packets and achieved a 99.6% MQTT packet delivery rate. During a rainfall-related disturbance, relative humidity increased from 54% to 86% while airflow fell to 0.1 m/s, and the system generated an alert within 2.1 seconds . These results demonstrate that the developed component was capable of detecting significant environmental changes and providing timely operational information.

The monitored drying process was reported as stabilising at 4.2 ± 0.3 days, compared with 6.5 ± 1.4 days under unmonitored ambient conditions. This result should be

interpreted as the combined effect of continuous monitoring and subsequent corrective actions rather than the effect of sensing alone. Nevertheless, the monitoring component played an important role by identifying unsafe environmental conditions early enough to support intervention.

Overall, the research demonstrates that a multi-parameter IoT architecture can provide reliable and continuous environmental visibility during Maldive fish drying. The developed component successfully combined sensing, edge processing, telemetry, threshold evaluation, and alerting within a single monitoring workflow and provided a practical foundation for further development of smart fish-processing systems.

8.2 Achievement of Objectives

The objectives established at the beginning of this research were addressed through the design, implementation, integration, and experimental evaluation of the developed monitoring component.

The first objective was to identify the key environmental parameters that influence Maldive fish drying. This was achieved through the selection of temperature, relative humidity, airflow, and atmospheric gas conditions as the main variables monitored by the system. These parameters were chosen because they collectively provide a broader representation of the drying microclimate than a single environmental measurement.

The second objective was to design and implement an ESP32-based multi-sensor IoT monitoring node. This objective was achieved through the integration of the DHT22, airflow sensor, and MQ-135 sensor with the ESP32 microcontroller. The resulting node provided the sensing and edge-processing functionality required for the individual component.

The third objective was to collect, process, and transmit environmental data for real-time monitoring. Environmental measurements were sampled every 30 seconds and processed using a one-minute sliding-window median filter. The validated telemetry was then transmitted through MQTT over Wi-Fi, enabling continuous monitoring of the drying environment.

The fourth objective was to implement a threshold-based environmental evaluation and alert mechanism. This was achieved through local ESP32 firmware that evaluated the interaction between relative humidity and airflow. A persistent high-humidity and low-airflow condition triggered an alert, allowing corrective action to be taken.

The fifth objective was to evaluate the technical performance of the monitoring component. This was achieved through continuous testing over a seven-day dehydration cycle. The 99.6% MQTT packet delivery rate, 10,080 validated telemetry packets, and 2.1-second alert response during the rainfall event provided direct evidence of technical performance.

Therefore, the research objectives were substantially achieved. The developed artefact successfully demonstrated the feasibility of continuous multi-parameter environmental monitoring and timely alert generation during Maldive fish drying.

8.3 Individual Research and Product Contributions

The principal individual research contribution of this study is the development and validation of a multi-parameter IoT-based environmental monitoring framework specifically designed for the drying stage of Maldive fish processing.

The research contribution lies in the integration of multiple environmental variables within a single embedded monitoring system. Instead of monitoring only temperature or humidity, the component combines temperature, relative humidity, airflow, and atmospheric gas information with edge-based processing, median filtering, MQTT telemetry, and threshold-based evaluation. This provides a more comprehensive representation of the drying environment.

A further research contribution is the use of combined-condition decision logic. The system does not evaluate humidity and airflow only as separate values. Instead, it identifies unsafe conditions when both variables simultaneously move outside the defined operating condition for a sustained period. This approach provides a more context-aware environmental assessment than a single fixed threshold.

The research also contributes empirical evidence from an extended monitoring period rather than relying only on short-duration laboratory testing. The seven-day

operational evaluation, communication reliability result, and response to the rainfall disturbance provide practical evidence that the proposed architecture can function under changing environmental conditions.

The individual product contribution is the functional ESP32-based IoT monitoring subsystem developed for integration into the wider Smart Maldive Fish Processing System. The prototype provides continuous environmental sensing, local data processing, MQTT communication, real-time monitoring, and alert support.

Another important product contribution is modularity. The environmental monitoring component remains technically separate from the raw fish assessment, boiling and salting optimisation, and final dried-fish assessment components. This means that future modifications to sensors, threshold rules, communication mechanisms, or monitoring interfaces can be introduced without redesigning the complete group system.

The component therefore contributes both a research artefact and a practical prototype that demonstrates how IoT technology can support the digitalisation of a traditional food-processing activity.

8.4 Future Directions

Although the developed monitoring component produced promising results, several areas remain for future improvement and investigation.

One important future direction is improved gas-sensor calibration and compensation. The MQ-135 sensor used in the current prototype is affected by environmental humidity, which may influence its response under changing drying conditions. Future work could introduce humidity-compensation algorithms, improved baseline calibration, or alternative sensing technologies to provide more stable gas measurements.

A second direction is longer-term and broader experimental validation. The current evaluation was conducted over a seven-day dehydration cycle. Future research should evaluate the system across multiple drying cycles, different seasons, weather

conditions, fish sizes, processing locations, and operational environments. This would provide stronger evidence regarding system robustness and generalisability.

Another important extension is the use of a distributed sensor network. The current monitoring approach is based on a localized sensing node, but larger drying environments may contain different microclimates. Multiple IoT nodes could be positioned at different locations within the drying area to capture spatial variations in temperature, humidity, airflow, and gas conditions.

Future development could also move the system from monitoring and alerting toward automated environmental control. Instead of notifying an operator only after detecting an adverse condition, future versions could automatically activate ventilation fans, adjust protective covers, or control other environmental mechanisms. Such automation could reduce response time further and support more consistent drying conditions.

A further research opportunity is the introduction of adaptive or predictive environmental intelligence. The current prototype uses deterministic threshold rules because they are transparent and easy to interpret. With a larger environmental dataset, future studies could investigate adaptive thresholds or predictive models capable of identifying developing risk conditions before the fixed threshold is exceeded.

Formal stakeholder and usability validation should also form part of future work. Fish-processing operators, supervisors, and other users could evaluate the clarity of dashboard information, usefulness of alerts, ease of operation, and perceived value of the monitoring system. This would provide evidence about practical acceptance in addition to the technical evidence produced in the current study.

Communication and system security could also be strengthened in future iterations. A commercial or large-scale deployment would require stronger authentication, encrypted communication, secure device management, fault recovery, and appropriate access controls.

Finally, future research could examine how environmental telemetry can be integrated more closely with information generated by the other components of the Smart Maldive Fish Processing System. Combining upstream processing data, drying-stage environmental information, and final product assessment could create a more complete

digital record of the production lifecycle and support improved traceability and process optimisation.

Overall, these future directions provide a pathway for extending the current research prototype into a more robust, scalable, automated, and commercially practical environmental monitoring solution for Maldive fish processing.

REFERENCES

[1] Y. He, C. Sun, H. Huang, L. Yang, X. Chen, and D. Tu, "Computer vision technology for food quality evaluation: A review," *Trends in Food Science & Technology*, vol. 62, pp. 152–166, 2017.

[2] J. Wang, J. Yue, Z. Zhou, K. Xie, L. Du, F. He, and Q. Yang, "Machine vision systems and applications for intelligent food quality inspection and evaluation: A review," *Food Engineering Reviews*, vol. 12, no. 4, pp. 542–567, 2020.

[3] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep learning in agriculture: A survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.

[4] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 6105–6114.

[5] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[6] T. Terven and D. Cordova-Esparza, "A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS," *arXiv preprint arXiv:2304.00501*, 2023.

[7] A. Yudhana, A. Djurdjani, and M. Kamal, "Fish freshness identification using machine learning: Performance comparison of k-NN and Naïve Bayes classifier," *Journal of Computer Science and Engineering*, vol. 16, no. 3, pp. 154–162, 2022.

[8] E. Prasetyo, N. Suciati, N. P. Sutramiani, A. Ananda, and A. P. W. K. Dewi, "DaFiF: A complete dataset for fish's freshness problems," *Data in Brief*, vol. 54, p. 110418, 2024.

[9] E. Prasetyo, N. Suciati, N. P. Sutramiani, A. Ananda, and A. P. W. K. Dewi, "Standardizing the fish freshness class during ice storage using clustering approach," *Ecological Informatics*, vol. 80, p. 102533, 2024.

[10] M. Kashani Zadeh, A. Alimardani, and M. Omid, "Rapid assessment of fish freshness using hyperspectral imaging and deep learning," *Sensors*, vol. 23, no. 5, p. 2736, 2023.

[11] H. Grassi, L. Palmieri, and G. Rizzotto, “Applications of computer vision in food quality evaluation: A review,” *Trends in Food Science & Technology*, vol. 118, pp. 150–165, 2022.

[12] S. M. R. Islam, M. Hasan, and M. A. Rahman, “Computer vision-based fish freshness classification using deep learning,” *IEEE Access*, vol. 11, pp. 58214–58229, 2023.

[13] A. Howard *et al*., “Searching for MobileNetV3,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1314–1324.

[14] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[15] M. S. Uddin, J. B. Alam, and S. M. M. Rahman, “Quality evaluation of dried fish products using digital image processing techniques,” *Journal of Food Process Engineering*, vol. 44, no. 6, p. e13689, 2021.

[16] S. Taheri and C. M. Lee, “Kinetic modeling of moisture loss and quality degradation during drying of marine products,” *Food Research International*, vol. 120, pp. 317–326, 2019.

[17] K. Wu, M. Debliquy, and C. Zhang, “Metal-oxide-semiconductor resistive gas sensors for fish freshness detection,” *Comprehensive Reviews in Food Science and Food Safety*, vol. 22, no. 2, pp. 913–945, 2023, doi: 10.1111/1541-4337.13095.

[18] S. A. O. Adeyeye, “An overview of fish drying kinetics,” *Nutrition & Food Science*, vol. 49, no. 5, pp. 886–902, 2019, doi: 10.1108/NFS-10-2018-0296.

GLOSSARY

Corrective Action – An operational response taken after the system detects an unsafe or undesirable environmental condition, such as improving airflow or protecting the drying area from adverse weather.

Drying Environment – The physical area in which processed Maldive fish is dehydrated and where surrounding conditions such as temperature, relative humidity, airflow, and atmospheric gases affect the drying process.

Drying Microclimate – The local environmental conditions immediately surrounding the fish during dehydration, which may differ from general outdoor weather conditions.

Environmental Monitoring – The continuous observation and measurement of environmental conditions that may influence the fish drying process.

Environmental Telemetry – Sensor-generated environmental data that are processed and transmitted digitally from the monitoring node to the wider monitoring system.

Median Filtering – A data-processing technique used to reduce short-term fluctuations or noise by selecting the median value from measurements within a defined time window.

Multi-Parameter Monitoring – The simultaneous observation of several environmental variables, such as temperature, relative humidity, airflow, and atmospheric gas conditions.

Operational Threshold – A predefined environmental condition used by the system to determine whether the drying environment requires attention or intervention.

Packet Delivery Rate – The proportion of transmitted telemetry packets that are successfully delivered through the communication system.

Sensor Recalibration – The process of adjusting a sensor baseline or response to maintain reliable measurements when environmental conditions change.

Telemetry Packet – A structured unit of processed environmental data prepared for transmission through the IoT communication system.

Threshold-Based Alerting – A method in which the system automatically generates a warning when monitored environmental conditions meet predefined unsafe criteria.

APPENDICES

Appendix A Research Artefacts and Extended Technical Evidence

This appendix provides supplementary technical evidence supporting the design, implementation, integration, and evaluation of the Intelligent IoT-Based Multi-Parameter Environmental Monitoring System for Fish Processing Optimization. The purpose of this appendix is to present additional research artefacts that strengthen the transparency and reproducibility of the individual study without repeating the essential discussion already provided in the main thesis chapters.

The materials included in this appendix should relate only to the individual environmental monitoring component. Relevant artefacts may include selected system architecture diagrams, sensor integration evidence, ESP32 firmware extracts, MQTT communication outputs, environmental telemetry samples, threshold configurations, testing screenshots, and additional records generated during the monitored drying period. These materials provide supporting evidence for the technical procedures described in Chapters 3, 4, 5, and 6.

The hardware-related evidence may include documentation showing how the ESP32 was connected to the environmental sensors used in the prototype. The individual component monitored temperature, relative humidity, airflow, and atmospheric gas conditions using a DHT22 sensor, an airflow sensing device, and an MQ-135 gas sensor. Supporting screenshots, wiring records, or configuration details may be included where available to demonstrate the implemented sensor setup.

The appendix may also contain selected firmware extracts showing the logic used for sensor acquisition, preprocessing, environmental condition evaluation, and telemetry preparation. These extracts should be limited to sections that directly demonstrate the implementation discussed in the main thesis. Large amounts of source code should not be included unless required as research evidence.

Communication-related artefacts may include selected MQTT transmission records or screenshots demonstrating the flow of telemetry from the ESP32 monitoring node to the wider monitoring environment. These artefacts can support the discussion of real-time data transmission and technical integration presented in Chapter 5.

Additional environmental monitoring records may also be included where available. These may show selected telemetry values, threshold-triggering events, or environmental changes observed during the drying process. Such evidence can provide further support for the validation results discussed in Chapter 6.

The appendix may further include selected testing outputs demonstrating that the monitoring node was able to operate continuously, process sensor data, transmit telemetry, and generate alerts when environmental conditions moved outside the defined operating range. These supporting materials should complement, rather than replace, the explanation and interpretation already presented in the main thesis.

Where artefacts from the wider Smart Maldive Fish Processing System are included, they should be clearly identified as integration evidence rather than as the individual contribution of this study. Results or technical outputs belonging specifically to raw fish assessment, boiling and salting optimisation, or final dried-fish quality assessment should not be presented as outcomes of the environmental monitoring component.

Overall, this appendix serves as a supporting technical record of the individual research process. It provides additional evidence of how the monitoring component was designed, implemented, connected, and validated while maintaining a clear distinction between essential thesis content and supplementary material.

Appendix B Academic Integrity and AI Use Disclosure

Generative Artificial Intelligence was used during the preparation of this thesis as a supporting tool for academic writing, language refinement, content organisation, and presentation improvement. The use of AI was limited to assisting with the communication and structuring of research material that originated from the researcher's own project documents, implementation work, presentations, research paper, technical evidence, and recorded experimental results.

The primary AI tool used was ChatGPT. It was used to support the transformation of project information into clearer academic prose, improve sentence structure, reduce grammatical errors, organise content according to the required thesis template, and refine explanations of technical workflows. AI support was also used to help structure

figure captions, glossary definitions, appendix descriptions, and some literature-review wording.

AI was not used as a substitute for the actual research process. The design of the system, hardware selection, sensor integration, ESP32 implementation, telemetry generation, threshold logic, testing activities, and experimental results were based on the project work and available research evidence. AI-generated suggestions were reviewed and retained only when they were consistent with the documented project materials.

All quantitative claims included in the thesis were checked against the available research evidence before inclusion. This included values related to sampling intervals, telemetry counts, MQTT packet delivery, threshold conditions, alert timing, humidity changes, airflow measurements, and reported drying-duration results. Unsupported accuracy values, participant numbers, experimental measurements, or implementation details were not intentionally introduced.

The researcher also reviewed AI-assisted literature-review content to ensure that references matched the claims being made. External literature was used only where relevant to the research topic, while project-specific claims were linked to the project's own research evidence. AI-assisted text was edited where necessary to maintain consistency with the terminology, scope, and findings of the study.

The use of AI also supported the preparation of visual descriptions and diagram structures. However, the technical meaning represented in these diagrams was based on the implemented project architecture and documented system workflow. AI-generated visuals were therefore treated as presentation aids rather than as independent research evidence.

Human review remained an essential part of the process. The researcher was responsible for checking the technical accuracy, relevance, originality, and appropriateness of all AI-assisted content before including it in the thesis. The final interpretation of findings, discussion of limitations, evaluation of contributions, and conclusions remained the responsibility of the researcher.

The researcher also remained responsible for compliance with academic integrity requirements. AI-generated wording was not treated as an authoritative source and was not cited as academic literature. Instead, academic claims were supported using the relevant scholarly references, while project-specific claims were based on the project's documented research outputs.